

THE MATHS BOX

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THE MATHS BOX

BOX 4

THE MATHS BOX CONTAINS:

- 1 Teachers Guide
- 75 maths activity cards
(2 of each card: 150 per box)
- 75 answer cards

• Number • Algebra • Shape and Space • Measures • Data

THE MATHS BOX

BOX 4

Features of The Maths Box:

- Colourful, creative and durable cards to encourage pupil motivation and enjoyment of maths.
- Ideal for either independent use or, as the cards are supplied in pairs, small-group and paired work.
- Written to ensure pupils can practise the skills, concepts and knowledge required by the curriculum.
- Designed to encourage pupils to apply their knowledge of mathematical concepts to a variety of high-interest activities and real-life maths situations.
- Suitable for all ability levels, with additional activities to challenge the more able pupil.

Features of the Teachers Guide:

- A scope and sequence chart, detailing the mathematical concept(s) covered by each card.
- Advice on how to use the box in the classroom.
- An explanation of the different parts of the activity cards.
- Answer recording sheets for pupils.
- Tracking and recording sheets for pupils and teachers.
- A list of the materials and equipment needed for each card.
- Full-colour posters to aid the teaching of card content and strategies.
- Photocopiable activity sheets to support, practise and extend the card activities.
- A full-colour copy of each answer card.
- A comprehensive glossary.

Benefits of The Maths Box:

- Encourages and motivates maths learning, giving pupils confidence in their maths skills.
- Connects maths to the real world through contextualised activities.
- Provides pupils with the skills, concepts and knowledge required by the curriculum.
- Helps pupils to become independent learners.
- Supports the teacher's maths syllabus with high-interest activities.
- Allows effective management of the teacher's time.

The Maths Box has six levels:

The Maths Box 1 CODE: 6987 ISBN: 978-1-84654-876-5
 The Maths Box 2 CODE: 6988 ISBN: 978-1-84654-877-2
 The Maths Box 3 CODE: 6989 ISBN: 978-1-84654-878-9
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Card 1



7 The building heights, from smallest to largest, are:
4.5 m, , 20 m

8 The bungalow has a height of 4.5 metres. How high is this in centimetres?
(a) 4500 cm (b) 450 cm (c) 45 cm

9 How much higher is the apartment block than the house?
(a) 11 m (b) 12 m (c) 13 m

10 Which building is exactly four times higher than the bungalow?
(a) office block (b) church (c) apartment block

11 If the bungalow was placed on top of the office block, they would almost be the same height as the

12 What is the total height of the five buildings?
(a) 62.5 m (b) 64.5 m (c) 74.5 m

13 True or False? 4.5 m can be written as $4\frac{1}{2}$ cm.

14 0.1 m = cm

ADDITIONAL Activity

- Draw a row of buildings. Use your ruler to make sure each building is the correct height. Your row of buildings should contain:
house (6 cm), house (7 cm), apartment block (17 cm), shop (5 cm)

Card 3

9 The temperature shown on the thermometer is:
(a) 45 °C
(b) 30 °C
(c) 40 °C

10 The full length of a tennis court is closer to:
(a) 5 m
(b) 25 m
(c) 55 m

11 Yes or No?
Is 50 metres a good estimate of the length of this swimming pool?

12 True or False?
The length of this water slide would be approximately 8 kilometres.

13 The difference in the level of water in the two jugs is:
(a) 30 mL
(b) 40 mL
(c) 50 mL

14 According to the set of luggage scales, the suitcase weighed:
(a) 20 kg
(b) 15 kg
(c) 25 kg

ADDITIONAL Activities

- Estimate, then use a ruler to measure the length and width of your desk. How close were your estimations?
- Select five objects in the classroom. Estimate the length and width of each object, then measure them using your ruler.

Card 2

6 How many kilograms of dog biscuits does the scales show?

7 The level of liquid in the jug is:
(a) 170 mL
(b) 160 mL
(c) 14.6 mL

8 If a toy from the bird's cage dropped into the jug and sank, the level of the water in the jug would:
(a) rise
(b) fall
(c) stay the same

Container	Capacity
bottle	750 mL
jug	400 mL
mug	200 mL
kettle	900 mL
bucket	2000 mL (2 litres)

- 9** What is the difference in capacity between the mug and the kettle?
- 10** How many times could you fill the jug with the bucket of water?
- 11** True or False?
1000 mL would spill from a full bucket of water if the full bottle was placed in it.
- 12** Which could not be used to measure capacity?
(a) a spoon
(b) a cup
(c) a tape measure
- 13** 100 grains of rice, 40 sunflower seeds and 15 pieces of macaroni all have a weight of 12 grams. The lightest would be one piece of
- 14** Which object would be about 15 cm long?
(a) an eraser
(b) a pen
(c) a ruler

ADDITIONAL Activities

- Research medical thermometers.
- What should a person's temperature normally be?
 - What are the lowest and highest temperatures shown on the scale of a medical thermometer?

Card 4

9 Ben's pet cat weighs 9.5 kg. How much is this in grams?
(a) 95 g
(b) 950 g
(c) 9500 g

10 Suzy has three bottles of different flavoured juices. The bottles contain 0.2 L, 0.3 L and 0.4 L. How many litres is this altogether? What is this in millilitres?
(a) 0.9 L, 900 mL
(b) 0.9 L, 90 mL
(c) 0.9 L, 9 mL

11 Jaz watched the TV for 240 minutes. How many hours is this?
(a) 3 (b) 3.5 (c) 4

12 Usain ran for 2500 metres. How many kilometres is this?

13 Jules had two bags to take on the plane as hand luggage. They weighed 5 kg and 9 kg. How many kilograms is this altogether? What is this in grams?

14 The caterpillar was 60 mm long. How many centimetres is this?

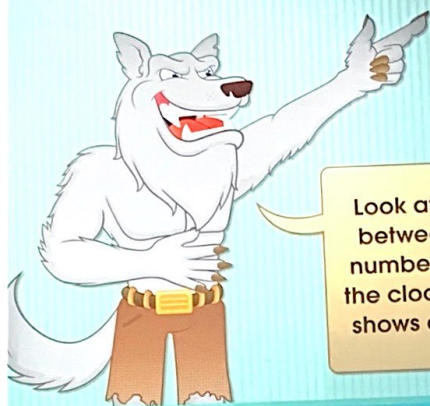
ADDITIONAL Activity

- Ben had three pet snakes. They measured 150 mm, 24 cm and 0.39 m. How long was each snake in centimetres? How many centimetres long would they be if they were placed nose to tail?

Card 1

Measures — Time

What's the Time, Mr Wolf?



Look at the four little marks between each five minute number, on the outer ring of the clock face. Each of these shows a time of one minute.

1 What time is it?

- (a) 7 minutes past 4
- (b) 17 minutes past 4
- (c) 7 minutes to 4



3 What time is it?

- (a) 36 minutes past 3
- (b) 36 minutes to 2
- (c) 36 minutes past 2



2 What time is it?

- (a) 9:24
- (b) 10:24
- (c) 11:24



4 What time is it?

- (a) 6:52
- (b) 7:52
- (c) 8:52



Card 3

Number — Fractions and Decimals

HUNDRETHS



Count up in hundredths. The missing fractions are:

1	2	3	4	6	7	9	10
100	100	100	100	100	100	100	100

- (a) $\frac{5}{100}$ and $\frac{8}{100}$
- (b) $\frac{15}{100}$ and $\frac{18}{100}$
- (c) $\frac{50}{100}$ and $\frac{80}{100}$

True or False?

An equivalent fraction for $\frac{10}{100}$ is $\frac{1}{10}$.

Count down in hundredths. The missing fractions are:

45	44	42	41	40	39	37	36
100	100	100	100	100	100	100	100

- (a) $\frac{43}{100}$ and $\frac{34}{100}$
- (b) $\frac{34}{100}$ and $\frac{83}{100}$
- (c) $\frac{43}{100}$ and $\frac{38}{100}$

True or False?

An equivalent fraction for $\frac{40}{100}$ is $\frac{5}{10}$.

5 Count up in hundredths. Record the missing fractions.

78	79	81	82	83	84	86	87
100	100	100	100	100	100	100	100

6 Count down in hundredths. Record the missing fractions.

63	61	60	59	58	57	56	54
100	100	100	100	100	100	100	100

7 The correct order of these fractions, from smallest to largest, is:

- (a) $\frac{98}{100}$, $\frac{71}{100}$, $\frac{32}{100}$, $\frac{10}{100}$, $\frac{7}{100}$
- (b) $\frac{10}{100}$, $\frac{32}{100}$, $\frac{7}{100}$, $\frac{71}{100}$, $\frac{98}{100}$
- (c) $\frac{7}{100}$, $\frac{10}{100}$, $\frac{32}{100}$, $\frac{71}{100}$, $\frac{98}{100}$

8 The correct order of these fractions, from smallest to largest, is:

- (a) $\frac{55}{100}$, $\frac{9}{10}$, $\frac{55}{100}$, $\frac{89}{100}$, $\frac{91}{100}$
- (b) $\frac{5}{10}$, $\frac{55}{100}$, $\frac{89}{100}$, $\frac{9}{10}$, $\frac{91}{100}$
- (c) $\frac{91}{100}$, $\frac{9}{10}$, $\frac{89}{100}$, $\frac{55}{100}$, $\frac{5}{10}$

Card 7

Number

Monster Numbers

WOW! These numbers are higher than 5000!

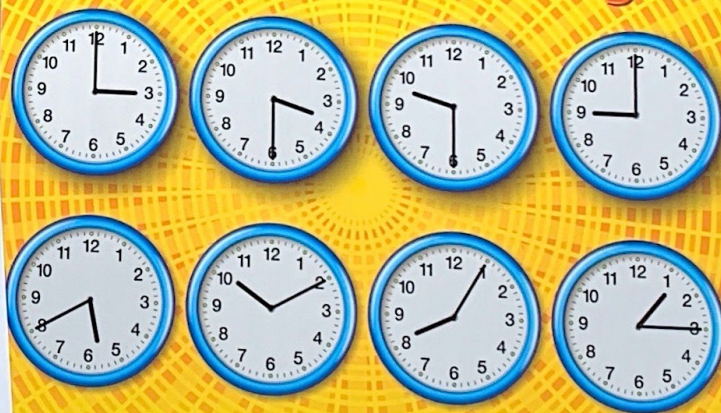


A	5121	5122	5123	5124		5126	5127	5128	5129	5130
B	5131	5132	5133	5134	5135	5136	5137	5138	5139	5140
C	5141	5142	5143	5144	5145		5147	5148	5149	5150
D	5151	5152	5153	5154	5155	5156	5157	5158	5159	5160
E	5161	5162	5163	5164	5165	5166	5167	5168	5169	5170
F	5171	5172	5173	5174	5175	5176	5177	5178	5179	5180
G	5181	5182	5183	5184	5185	5186	5187	5188	5189	5190
H	5191	5192	5193	5194	5195	5196	5197	5198	5199	5200
I	5201	5202	5203	5204	5205	5206	5207	5208	5209	5210
J	5211	5212	5213	5214	5215	5216	5217	5218	5219	5220
K	5221	5222	5223	5224	5225	5226	5227	5228	5229	5230
L		5232	5233	5234	5235	5236	5237	5238	5239	5240

Card 10

Shape and Space

Clock Hands and Angles



1 Look at the times shown on the clocks. At which times do the minute and hour hands make right angles?

- (a) 5:40 and 10:10
- (b) 3:30 and 9:30
- (c) 3:00 and 9:00

2 At 3:30, the smaller angle made by the hour and minute hands is:

- (a) greater than a right angle
- (b) less than a right angle
- (c) a right angle

3 At 8:05, the smaller angle made by the hour and minute hands is:

- (a) an acute angle
- (b) an obtuse angle
- (c) a right angle

4 Compare the smaller of the two angles made by the hour and minute hands at the three times below. This angle is **not** acute at:

- (a) 5:40
- (b) 3:30
- (c) 8:05

Card 1



5 How many more minutes until 10:00?



6 How many minute since 1:00?



7 The digital time that Mr Wolf has breakfast is :



8 What time does Mr Wolf eat his lunch?
(a) 12 minutes to 12
(b) 12 minutes past 12
(c) 21 minutes past 12



9 True or False?
Mr Wolf eats his dinner at 7:57.

10 True or False?
Mr Wolf ate his breakfast in the morning, before noon.

11 True or False?
Looking at the clock above, there are three more minutes until 7 o'clock.

12 True or False?
The closest digital time to 8:00 on the clocks on this side of the card is 9:19.

13 What time is two hours earlier than the time shown on the clock above?

14 What time is three hours later than the time shown on the clock above?

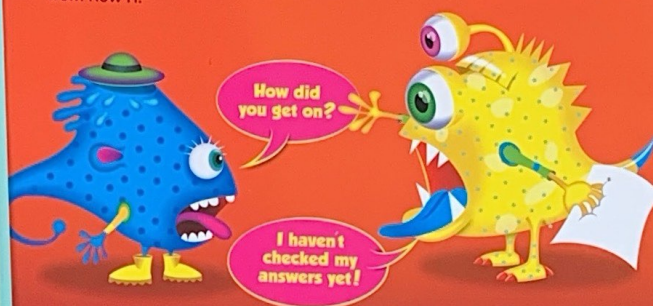
ADDITIONAL Activity

Draw five clock faces, showing the following times:

- 4 o'clock
- 6:15
- 23 minutes past 10
- quarter to 11
- 2:39 p.m.

Card 7

- What number is missing from Row A?
(a) 6125 (b) 5125 (c) 5124
 - The number is missing from Row C.
 - Are the numbers in the green column odd or even?
 - Are the numbers in the yellow column odd or even?
- Fill in the missing sign: < or >
- 5183 > 5173
 - 5200 < 5120
 - The number is missing from Row H.
 - What number is 10 more than 5201?
(a) 5211 (b) 5202 (c) 5210
 - What number is 10 more than 5186?
 - What number is 20 less than 5176?
(a) 5196 (b) 5156 (c) 5126
 - What number is 20 less than 5232?
 - The number is missing from Row L.
 - What number would be 100 less than the lowest number on the chart?
 - Change around the digits in the highest number on the chart to make the lowest possible four-digit number.



ADDITIONAL Activities

Write all the even numbers in Row B and Row D.
Starting at 5240, count backwards in 5s until you reach 5205. Write each number.

Card 3

Hundredths arise when dividing any units number by one hundred.
For example: $4 \div 100 = 0.04$ or $\frac{4}{100}$
The numbers in this table have all been divided by one hundred.

NUMBER	HUNDREDS	TENS	UNITS	TENTHS $\frac{1}{10}$	HUNDREDTHS $\frac{1}{100}$
4				0	4
57				5	7
129			1	2	9
8395	8	3		9	5

- 9 True or False?
 $63 \div 100 = 0.63$ or $\frac{63}{100}$
- 10 True or False?
 $5 \div 100 = 0.05$ or $\frac{5}{100}$
- 11 $24 \div 100 = 0.....$ or $\frac{24}{100}$

Hundredths also arise when dividing tenths by ten.
For example: $0.2 \div 10 = 0.02$ or $\frac{2}{100}$
The numbers in this table have all been divided by ten.

NUMBER	HUNDREDS	TENS	UNITS	TENTHS $\frac{1}{10}$	HUNDREDTHS $\frac{1}{100}$
0.3				0	3
4.5				4	5
43.9			4	3	9
170.2	1	7		0	2

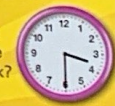
- 12 True or False?
 $0.5 \div 10 = 0.05$ or $\frac{5}{100}$
- 13 True or False?
 $7.8 \div 10 = 7.8$ or $7\frac{8}{100}$
- 14 $6.3 \div 10 = 0.....$ or $\frac{63}{100}$

ADDITIONAL Activity

Draw a table like the one above. Choose four numbers to write in the 'Number' column. Divide each number by 100 and write the answer in the correct place value columns.

Card 10

- 5 True or False?
The angle made by the clock hands at 3:00 is greater than the angle made at 9:00.
- 6 When the minute hand goes completely around the clock, it makes:
(a) a straight angle
(b) an obtuse angle
(c) a full turn
- 7 If it is 9 o'clock, what time would it be if the minute hand moved one full right angle?
(a) 9:15 (b) 9:30 (c) 9:45
- 8 At 1:15, the smaller angle made by the hour and minute hands is:
(a) an obtuse angle
(b) an acute angle
(c) a straight angle
- 9 At which time on the clock is the angle between the minute and hour hands a straight angle?
(a) 2:45 (b) 6:00 (c) 9:15
- 10 Which time makes an acute angle?
(a) 3 o'clock
(b) 6 o'clock
(c) 10 o'clock
- 11 True or False?
The smaller angle in Clock A < the smaller angle in Clock B
- 12 What type of angle is the smaller angle shown on this clock?
- 13 Which two times make obtuse angles?
(a) 2 o'clock and 11 o'clock
(b) 3 o'clock and 9 o'clock
(c) 5 o'clock and 8 o'clock
- 14 Which two times make acute angles?
(a) 2 o'clock and 11 o'clock
(b) 4 o'clock and 9 o'clock
(c) 5 o'clock and 8 o'clock

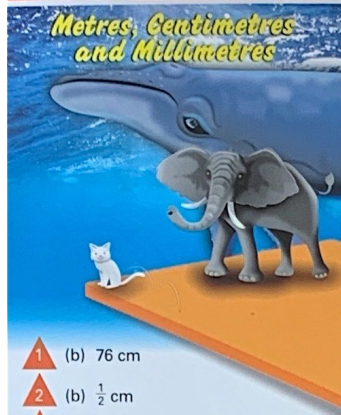


ADDITIONAL Activities

- Find images of angles in the natural environment; e.g. between a tree's branches and its trunk, the antennae of an insect, the bend of a frog's legs, the veins of a leaf.
- Choose an image to copy carefully.
- Describe the types of angles your picture shows.

Card 1

Answers

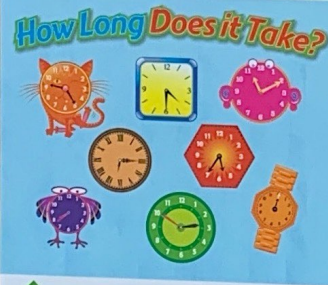


- 1 (b) 76 cm
- 2 (b) $\frac{1}{2}$ cm
- 3 (a) 6 m
- 4 (c) 3 m
- 5 (a) 60
- 6 True
- 7 7 m, 15 m, 18 m
- 8 (b) 450 cm
- 9 (c) 13 m
- 10 (b) church
- 11 apartment block
- 12 (b) 64.5 m
- 13 False
- 14 10

Card 6

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Answers



- 1 (b) 1 h 45 min
- 2 True
- 3 (a) 3 h 35 min
- 4 E, 2:50 p.m.
- 5 (a) Clock H
- 6 (a) Clock G and Clock L
- 7 Clock L
- 8 (c) 4 h 15 min
- 9 (b) Clock K
- 10 Clock I, 3:45
- 11 Five twenty-six
- 12 (a) 3:00 p.m. to 4:30 p.m.
- 13 Movie A
- 14 (c) 285 minutes

Card 1

Answers

Which Shape am I?



- 1 Octagon
- 2 Triangle
- 3 Rhombus
- 4 Hexagon
- 5 Pentagon
- 6 Equilateral triangle
- 7 Scalene triangle
- 8 Isosceles triangle
- 9 Right-angled triangle
- 10 Rectangle
- 11 Parallelogram
- 12 Kite
- 13 True
- 14 False

Card 1

Answers

WHAT ARE THE CHANCES?



- 1 (c) low temperature and ice
- 2 (b) snow and drought
- 3 (b) boots and gloves
- 4 (c) the temperature inside
- 5 (b) flood
- 6 (c) drought
- 7 (a) destructive winds
- 8 (a) ice skating and ice hockey
- 9 (b) bought a ticket
- 10 (a) many tickets were sold
- 11 (c) started well
- 12 (b) leave home early enough
- 13 (b) tell you to do it at lunchtime
- 14 (c) The dog ate it

Card 1

Answers

Magic Number Squares



- 1 (b) 12
- 2 15
- 3 (a) 5
- 4 4
- 5 34
- 6 (a) 7
- 7 (c) 3
- 8 (a) the letter C
- 9 (a) A
- 10 The letter Y
- 11 True
- 12 (c) 7
- 13 15
- 14 (c) by adding

Card 2

Answers

PET PARADE



- 1 (a) 48 kilograms
- 2 (c) 6 kg
- 3 (a) 1.2 metres
- 4 True
- 5 True
- 6 3.6 kg
- 7 (a) 170 mL
- 8 (a) rise
- 9 700 mL
- 10 5 times
- 11 False
- 12 (c) a tape measure
- 13 rice
- 14 (b) a pen